

Electron tube --- Solution

Y25 (2025) — English translation

A1 Let the electron begin to move with zero initial speed from the point C . Show that in this case $x(t) = 0$.

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At the initial moment, the electron acquires speed along the Oy axis, along the electric field. This leads to the emergence of a magnetic force along Oz and the appearance of a velocity component v_z . The magnetic force associated with v_z is parallel to the Oy axis. Thus, in the process of further movement, the forces acting on the electron lie in the Oyz plane, from which follows $x(t) = 0$.

A2 For the described case, write down Newton's second law in projection on the y, z axes, from which you will obtain expressions for $\ddot{y}, \ddot{z}$. Expressions may contain $\dot{y}, \dot{z}, y, z, U, I$, as well as geometric characteristics of the system.

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In a flat configuration, the electric field is uniform and equal to

$$E_y = \frac{\varphi_K - \varphi_A}{h} = -\frac{U}{h}.$$

Магнитное поле провода при движении электрона в плоскости Oyz параллельно Ox равно (например, из теоремы о циркуляции)

$$B_x = -\frac{\mu_0 I}{2\pi(y + r_0)}.$$

С учётом отрицательности знака заряда электрона получаем уравнения движения:

$$\begin{cases} m\ddot{y} = -eE_y - e\dot{z}B_x \\ m\ddot{z} = e\dot{y}B_x \end{cases}$$

Substituting expressions for the fields, we get the answer.

Answer:

$$\begin{cases} \ddot{y} = \frac{eU}{mh} + \frac{\mu_0 e I}{2\pi m(y + r_0)} \dot{z} & (1) \\ \ddot{z} = -\frac{\mu_0 e I}{2\pi m(y + r_0)} \dot{y} & (2) \end{cases}$$

A3 Obtain a differential equation for y of the form

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$$\ddot{y} = \alpha - \beta \frac{\ln\left(1 + \frac{y}{r_0}\right)}{1 + \frac{y}{r_0}},$$

где α, β – константы. Найдите α, β . The answer may contain all previously entered quantities and constants.

Multiplying the second equation by dt , we arrive at the relation

$$dz = -\frac{\mu_0 e I}{2\pi m(y + r_0)} dy.$$

Интегрируем:

$$\dot{z} = -\frac{\mu_0 e I}{2\pi m} \int \frac{dy'}{y' + r_0} = -\frac{\mu_0 e I}{2\pi m} \ln\left(1 + \frac{y}{r_0}\right) + C.$$

С учётом начальных условий $\dot{z} = 0$ при $y = 0$, тогда $C = 0$, и

$$\dot{z} = -\frac{\mu_0 e I}{2\pi m} \ln\left(1 + \frac{y}{r_0}\right).$$

Постановка полученного выражения в (1) даёт

$$\ddot{y} = \frac{eU}{mh} - \frac{1}{r_0} \left(\frac{\mu_0 e I}{2\pi m}\right)^2 \frac{\ln\left(1 + \frac{y}{r_0}\right)}{1 + \frac{y}{r_0}}. \quad (3)$$

From here we get the answer.

Answer:

$$\alpha = \frac{eU}{mh} \quad \beta = \frac{1}{r_0} \left(\frac{\mu_0 e I}{2\pi m}\right)^2$$

After multiplying by $\dot{y}$, the equation (3) is transformed to the form

$$\frac{\dot{y}^2}{2} = \int \left(\frac{eU}{mh} - \frac{1}{r_0} \left(\frac{\mu_0 e I}{2\pi m} \right)^2 \frac{\ln \left(1 + \frac{y}{r_0} \right)}{1 + \frac{y}{r_0}} \right) dy.$$

Интеграл во втором слагаемом:

$$\int \frac{\ln \left(1 + \frac{y}{r_0} \right)}{1 + \frac{y}{r_0}} dy = r_0 \int \ln \left(1 + \frac{y}{r_0} \right) d \left(\ln \left(1 + \frac{y}{r_0} \right) \right) = \frac{r_0}{2} \ln^2 \left(1 + \frac{y}{r_0} \right) + C.$$

С учётом начальных условий $\dot{y} = 0$ при $y = 0$, тогда $C = 0$, и

$$\frac{\dot{y}^2}{2} = \frac{eUy}{mh} - \frac{1}{2} \left(\frac{\mu_0 e I}{2\pi m} \right)^2 \ln^2 \left(1 + \frac{y}{r_0} \right).$$

Условие достижения электроном анода:

$$\dot{y}(h) = 0, 0 = \frac{eU}{m} - \frac{1}{2} \left(\frac{\mu_0 e I_{kp}}{2\pi m} \right)^2 \ln^2 \left(1 + \frac{h}{r_0} \right).$$

Отсюда получаем ответ для I_{cr} .

Answer:

$$I_{kp} = \sqrt{\frac{8Um}{e}} \frac{\pi}{\mu_0} \frac{1}{\ln \left(1 + \frac{h}{r_0} \right)}$$

A5 Let the electron start moving with zero initial speed from the point C in the cylindrical lamp configuration. In this case, find the maximum value of the heater current I_{kp} at which the electron reaches the anode. 1.20

Similar to the previous situation, it is shown that the movement will occur in the Oyz plane if the axes are introduced in the same way as in the figure for a plane configuration.

Method 1

We will consider the movement in coordinates (r, z) , where r is the distance from the system axis to the point. Let's calculate the electric field using Gauss's theorem. Due to the absence of volumetric charges

$$E_r \cdot r = \text{const.} \Rightarrow E_r = \frac{A}{r}, U = - \int_{r_0}^{R_0} E_r dr = -A \ln \left(\frac{R_0}{r_0} \right), E_r = - \frac{U}{r \ln \left(\frac{R_0}{r_0} \right)}.$$

Магнитное поле:

$$B_x = - \frac{\mu_0 I}{2\pi r}.$$

Тогда уравнения движения выглядят следующим образом:

$$\begin{cases} \ddot{r} = - \frac{eU}{mr \ln \left(\frac{R_0}{r_0} \right)} + \frac{\mu_0 e I}{2\pi m(y + r_0)} \dot{z} \\ \ddot{z} = - \frac{\mu_0 e I}{2\pi m(y + r_0)} \dot{r} \end{cases}$$

Аналогичным образом после интегрирования второго уравнения, подстановки в первое и повторного интегрирования получаем уравнение на I_{cr} :

$$\frac{eU}{m} - \frac{1}{2} \left(\frac{\mu_0 e I_{kp}}{2\pi m} \right)^2 \ln^2 \left(\frac{R_0}{r_0} \right).$$

From here we get the answer.

Method 2

Let's consider the equations of motion from point A2, without substituting the specific values of E_y and B_x :

$$\begin{cases} \ddot{y} = - \frac{e}{m} (E_y(y) + \dot{z} B_x(y)) \\ \ddot{z} = \frac{e}{m} \dot{y} B_x(y) \end{cases}.$$

С учётом начальных условий из второго уравнения получим:

$$\dot{z} = \frac{e}{m} \int_0^y B_x(\xi) d\xi.$$

Подставим в первое уравнение:

$$\ddot{y} = -\frac{e}{m} \left(E_y(y) + \frac{e}{m} B_x(y) \int_0^y B_x(\xi) d\xi \right).$$

Домножим на $\dot{y}$ и проинтегрируем с учётом начальных условий:

$$\frac{\dot{y}^2}{2} = -\frac{e}{m} \int_0^y E_y(\eta) d\eta - \frac{e^2}{m^2} \int_0^y B_x(\eta) d\eta \int_0^\eta B_x(\xi) d\xi = -\frac{e}{m} (\varphi(y) - \varphi(0)) - \frac{e^2}{m^2} \int_0^y B_x(\eta) d\eta \int_0^\eta B_x(\xi) d\xi. \quad \text{[[10]]} \quad \frac{e}{m} \int_0^h B_x(\eta) d\eta \int_0^\eta B_x(\xi) d\xi = \varphi(y) -$$

Таким образом, условие не зависит от распределения поля E_y , но только от разности потенциалов U . Так как распределение магнитного поля в цилиндрической и в плоской системе совпадают, то и ответ для I_{cr} (с точностью до подстановки $R_0 = r_0 + h$) remains the same.

Note. It can be shown that

$$\int_0^h B_x(\eta) d\eta \int_0^\eta B_x(\xi) d\xi = \frac{1}{2} \left(\int_0^h B_x(\xi) d\xi \right)^2.$$

Таким образом, условие на достижение электроном анода:

$$\int_0^h B_x(\xi) d\xi = \sqrt{\frac{2mU}{e}}.$$

Интересный факт заключается в том, что найденное условие зависит только от значений определённых интегралов электрического и магнитного поля по y , but not from a specific field distribution.

It can be shown that

$$\int_0^h B_x(\eta) d\eta \int_0^\eta B_x(\xi) d\xi = \frac{1}{2} \left(\int_0^h B_x(\xi) d\xi \right)^2.$$

Thus, the condition for the electron to reach the anode is:

$$\int_0^h B_x(\xi) d\xi = \sqrt{\frac{2mU}{e}}.$$

An interesting fact is that the found condition depends only on the values of certain integrals of the electric and magnetic field over y , but not on the specific field distribution.

Answer:

$$I_{kp} = \sqrt{\frac{8Um}{e}} \frac{\pi}{\mu_0} \frac{1}{\ln\left(\frac{R_0}{r_0}\right)}$$

B1 Derive a differential equation relating $\varphi(y)$ and $\rho(y)$.

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Method 1

Let's write Gauss's theorem for two planes y and $y + dy$:

$$(E(y + dy) - E(y))S = \frac{\rho(y)}{\varepsilon_0} S dy, \quad E'(y) = \frac{\rho(y)}{\varepsilon_0}.$$

С учётом связи

$$E(y) = -\varphi'(y)$$

we get the answer.

Method 2

Gauss's theorem in differential form:

$$\frac{\rho}{\varepsilon_0} = \operatorname{div} \vec{E}.$$

Связь E и φ :

$$E = -\operatorname{grad} \varphi.$$

Тогда

$$\frac{\rho}{\varepsilon_0} = -\operatorname{div} \operatorname{grad} \varphi = -\Delta \varphi,$$

здесь $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplace operator (Laplacian), the equation itself is called the Laplace equation. In the one-dimensional case this transforms to the form

Answer:

$$\varphi''(y) = -\frac{\rho(y)}{\varepsilon_0}$$

B2 Let's assume that $U > 0$. Find the speed of the electron $v(y)$ emitted from the cathode at a distance y from the cathode. Express your answer using $\varphi(y)$.

0.20

Let us write down the law of conservation of energy for the electron:

$$\frac{mv^2(y)}{2} - e\varphi(y) = 0.$$

From this follows the answer.

Answer:

$$v = \sqrt{\frac{2e\varphi(y)}{m}}$$

B3 Express the total current I flowing through the section $y = \text{const}$. Express your answer in terms of $\rho(y)$, $v(y)$ and the geometric characteristics of the system. Consider the direction of current from the anode to the cathode to be positive.

0.30

Taking into account the indicated positive direction of current from the anode to the cathode

Answer:

$$I = -\rho(y)v(y)S$$

B4 Obtain a differential equation on $\varphi(y)$ containing the function $\varphi(y)$ and its derivatives, as well as I and the geometric characteristics of the system.

0.40

Taking into account the previous points:

$$\rho(y) = -\frac{I}{Sv(y)} = -\frac{I}{S} \sqrt{\frac{m}{2e\varphi(y)}}.$$

Постановка $\rho(y)$ into the equation of point B1 leads to the answer

Answer:

$$\varphi''(y) = \frac{I}{\varepsilon_0 S} \sqrt{\frac{m}{2e\varphi(y)}}$$

B5 Get addicted $\varphi(y)$. Express your answer in terms of I and the geometric characteristics of the system.

0.80

Multiplying both sides of the equation of item B4 by $\varphi'(y)$ and integrating taking into account $\varphi'(0) = 0$, we arrive at the equation

$$(\varphi')^2 = \frac{I}{\varepsilon_0 S} \sqrt{\frac{8m}{e}} \sqrt{\varphi}.$$

Переменные в нём разделяются:

$$\varphi^{-1/4} d\varphi = \left(\frac{I}{\varepsilon_0 S} \right)^{1/2} \left(\frac{8m}{e} \right)^{1/4} dy.$$

Применяя начальное условие $\varphi(0) = 0$, получим

$$\frac{4}{3} \varphi^{3/4} = \left(\frac{I}{\varepsilon_0 S} \right)^{1/2} \left(\frac{8m}{e} \right)^{1/4} y.$$

Transforming the resulting expression, we arrive at the answer.

$$(\varphi')^2 = \frac{I}{\varepsilon_0 S} \sqrt{\frac{8m}{e}} \sqrt{\varphi}.$$

The variables in it are separated:

$$\varphi^{-1/4} d\varphi = \left(\frac{I}{\varepsilon_0 S} \right)^{1/2} \left(\frac{8m}{e} \right)^{1/4} dy.$$

Applying the initial condition $\varphi(0) = 0$, we obtain

$$\frac{4}{3} \varphi^{3/4} = \left(\frac{I}{\varepsilon_0 S} \right)^{1/2} \left(\frac{8m}{e} \right)^{1/4} y.$$

Transforming the resulting expression, we arrive at the answer.

Answer:

$$\varphi(y) = \left(\frac{3y}{4} \right)^{4/3} \left(\frac{I}{\varepsilon_0 S} \right)^{2/3} \left(\frac{8m}{e} \right)^{1/3}$$

B6 The current-voltage characteristic of the lamp at $U > 0$ can be expressed by the function

0.80

$$I(U) = GU^\gamma.$$

Определите коэффициенты G, γ . Ответ выразите через геометрические характеристики системы. Величина G is called the perveance of the lamp.

The value G is called the lamp perveance.

The voltage across the lamp is equal to the potential at point $y = h$:

$$U = \varphi(h) = \left(\frac{3h}{4} \right)^{4/3} \left(\frac{I}{\varepsilon_0 S} \right)^{2/3} \left(\frac{8m}{e} \right)^{1/3}.$$

Выразим ток:

$$I(U) = \varepsilon_0 S \left(\frac{4}{3h} \right)^2 \left(\frac{e}{8m} \right)^{1/2} U^{3/2}.$$

Then

Answer:

$$G = \varepsilon_0 S \left(\frac{4}{3h} \right)^2 \left(\frac{e}{8m} \right)^{1/2}$$

$$\gamma = \frac{3}{2}$$

B7 Get addicted $\rho(y)$. Express your answer in terms of I and the geometric characteristics of the system.

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Using the results of points B1 or B4, we express $\rho(y)$ in terms of $\varphi(y)$:

$$\rho(y) = -\varepsilon_0 \cdot \varphi''(y) = -\frac{I}{S} \sqrt{\frac{m}{2e\varphi(y)}}.$$

Подставляя найденное $\varphi(y)$, we arrive at the answer

Answer:

$$\rho(y) = -\left(\frac{I}{3S} \right)^{2/3} \left(\frac{2m\varepsilon_0}{e} \right)^{1/3} \cdot y^{-2/3}$$

B8 Obtain the current-voltage characteristic of lamp $I(U)$ at $U < 0$.

0.20

At $U < 0$ the field is directed along the y axis, and the electrons emitted from the cathode do not move towards the anode, which means $I = 0$.

Answer:

$$I(U) = 0$$

B9 Obtain a differential equation for $\varphi(r)$ containing the function $\varphi(r)$ and its derivatives, as well as the total current in the lamp I and the geometric characteristics of the system.

0.60

The current flowing in a cylindrical configuration is

$$I = -2\pi r L \rho(r) v(r).$$

С учётом выражения для скорости получаем

$$\rho(r) = -\frac{I}{2\pi r L} \sqrt{\frac{m}{2e\varphi(r)}}$$

Method 1

Consider cylindrical surfaces of length L and radii r and $r + dr$. Gauss's theorem in this case will be written in the form:

$$2\pi L (E(r + dr)(r + dr) - E(r)r) = \frac{2\pi L r \rho(r) dr}{\varepsilon_0}, \quad \frac{1}{r} \frac{d}{dr} (rE(r)) = \frac{\rho(r)}{\varepsilon_0} = -\frac{I}{2\pi r L \varepsilon_0} \sqrt{\frac{m}{2e\varphi(r)}}.$$

Используя $E(r) = -\varphi(r)$, we arrive at the equation for the potential.

Method 2

In method 2 of point B1,

$$\Delta\varphi = -\frac{\rho}{\varepsilon_0}.$$

Оператора Лапласа в цилиндрических координатах:

$$\Delta\varphi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \theta^2} + \frac{\partial^2 \varphi}{\partial z^2}.$$

В силу осевой симметрии системы и $E_z = 0$ последние два слагаемых равны нулю. После подстановки ρ через I was obtained, we get the answer.

Answer:

$$r\varphi''(r) + \varphi'(r) = \frac{I}{2\pi L \varepsilon_0} \sqrt{\frac{m}{2e\varphi(r)}}$$

C1 Kirchhoff's law for the RLC circuit leads to a differential equation of the form

0.90

$$\ddot{U}_C + \zeta(1 + \eta U_C^2) \dot{U}_C + \chi U_C = 0.$$

Выразите ζ, η, χ через R, L, C, M, λ, μ .

The current in the RLC circuit is related to the capacitor voltage:

$$I = C \dot{U}_C.$$

Тогда для RLC-контура записывается закон Кирхгофа:

$$M \frac{dI_A}{dt} = U_C + RC \dot{U}_C + LC \ddot{U}_C \quad (1)$$

(с точностью до знака M). Используя выражение для $I_A(U_C)$, получим

$$\frac{dI_A}{dt} = (\lambda - 3\mu U_C^2) \dot{U}_C. \quad (2)$$

После подстановки (2) в (1) и сокращения на LC , приходим к уравнению

$$\ddot{U}_C + \left(\frac{R}{L} - \frac{\lambda M}{LC} + \frac{3\mu M}{LC} U_C^2 \right) \dot{U}_C + \frac{U_C}{LC} = 0.$$

Comparing it with the data in the condition, we get the answer.

Answer:

$$\zeta = \frac{R}{L} - \frac{\lambda M}{LC}, \quad \eta = \frac{3\mu M}{RC - \lambda M}, \quad \chi = \frac{1}{LC}$$

C2 Let U_C deviate from zero by a small amount. Obtain a condition for the parameters ζ, η, χ under which the oscillations build up. Also express the resulting condition in terms of R, L, C, M, λ, μ .

1.20

For small U_C we can neglect the $\sim U_C^2$ term in the equation and write it in the form:

$$\ddot{U}_C + \zeta \dot{U}_C + \chi U_C = 0.$$

Method 1 (mathematical)

Let's introduce the notation:

$$\zeta = 2\gamma, \quad \chi = \omega_0^2.$$

Как известно, решение такого дифференциального уравнения имеет разный вид в зависимости от параметров γ, ω_0 . $\gamma^2 < \omega_0^2$

$$U_C(t) = e^{-\gamma t} \left(A \cos \left(\sqrt{\omega_0^2 - \gamma^2} \cdot t \right) + B \sin \left(\sqrt{\omega_0^2 - \gamma^2} \cdot t \right) \right).$$

$$\gamma^2 = \omega_0^2$$

$$U_C(t) = e^{-\gamma t} (A + Bt).$$

$$\gamma^2 > \omega_0^2$$

$$U_C(t) = e^{-\gamma t} \left(A \exp \left(\sqrt{\gamma^2 - \omega_0^2} \cdot t \right) + B \exp \left(-\sqrt{\gamma^2 - \omega_0^2} \cdot t \right) \right).$$

Второй случай не имеет физического смысла, так как требуется точное выполнение условия $\gamma^2 = \omega_0^2$. В остальных случаях, как видно, раскачка возможна только при условии $\gamma < 0$, то есть $\zeta < 0$.

As is known, the solution to such a differential equation has a different form depending on the parameters γ, ω_0 .

The second case has no physical meaning, since the exact fulfillment of the condition $\gamma^2 = \omega_0^2$ is required. In other cases, as can be seen, swinging is possible only under the condition $\gamma < 0$, that is, $\zeta < 0$.

In the third case, this does not lead to exponential growth of U_C , since the resulting linear equation is valid only for small U_C . A characteristic phase diagram of the exact solution is shown in the figure.

Method 2 (energy)

Energy in the RLC circuit:

$$W = \frac{CU_C^2}{2} + \frac{LC^2\dot{U}_C^2}{2}.$$

Продифференцируем по времени:

$$\dot{W} = CU_C\dot{U}_C + LC^2\dot{U}_C\ddot{U}_C = C\dot{U}_C(U_C + LC\ddot{U}_C).$$

Подставим $\ddot{U}_C$ из исходного уравнения, выразив его через $\dot{U}_C, U_C$. После преобразований приходим к выражению

$$\dot{W} = C\dot{U}_C^2(M\lambda - RC - 3\mu MU_C^2).$$

При малых U_C :

$$\dot{W} \simeq C\dot{U}_C^2(M\lambda - RC).$$

Для раскачки колебаний необходимо $\dot{W} > 0$, which is equivalent

Answer:

$$\zeta < 0 \quad RC < \lambda M$$

